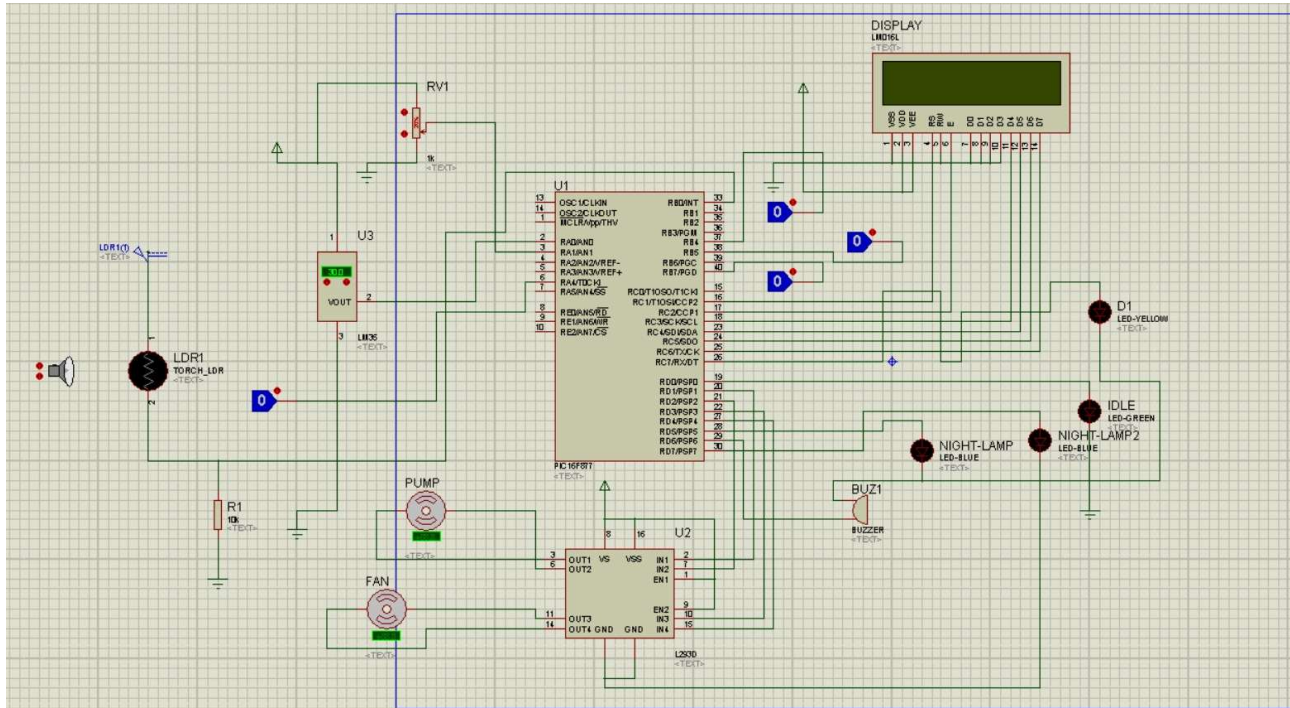


Team 4:
 Chaima Khayati
 Iheb Hamouda
 Ahmed ben khelifa
 Tasnim Guesmi
 Ahmed Safierrahman Abidi
 Fatma Zahra Joobeur



```
// =====
//   LCD PIN CONFIG
// =====
```

```
sbit LCD_RS at RC1_bit;
sbit LCD_EN at RC2_bit;
sbit LCD_D4 at RC3_bit;
sbit LCD_D5 at RC4_bit;
sbit LCD_D6 at RC5_bit;
sbit LCD_D7 at RC6_bit;
```

```
sbit LCD_RS_Direction at TRISC1_bit;
sbit LCD_EN_Direction at TRISC2_bit;
sbit LCD_D4_Direction at TRISC3_bit;
sbit LCD_D5_Direction at TRISC4_bit;
sbit LCD_D6_Direction at TRISC5_bit;
sbit LCD_D7_Direction at TRISC6_bit;
```

```
int i, j;
void interrupt() {

    if (INTCON.T0IF==1){
        i=1;
        TMR0=253;
        INTCON.T0IF=0;
```

```

    }

}

// =====
//     IDLE STATE
// =====
void idle_state() {

    PORTD.RD0 = 1;  // LED ON
    Lcd_Cmd(_LCD_CLEAR);
    // Fan, heater, pump, lighting OFF
    PORTB.RB0 = 0;
    PORTD.RD1 = 0;
    PORTD.RD2 = 0;
    PORTD.RD3 = 0;
    PORTD.RD4 = 0;
    PORTE.RE2 = 0;

    Lcd_Out(1,1, "AQUARIUM CONTROL");
    Lcd_Out(2,1, "SYSTEM READY  ");
}

// =====
//     MAIN
// =====
void main() {

    // Interrupt configuration
    INTCON.GIE = 1;
    INTCON.INTE = 1;
    INTCON.RBIE = 1;

    intcon.T0IE=1;
    OPTION_REG.T0CS=1;
    OPTION_REG.T0SE=1;
    TMR0=253;

    // I/O configuration
    TRISA = 0xFF;
    TRISB = 0b10110001;
    TRISC = 0x00;
    TRISD = 0x00;
    TRISE = 0x00;

    ADC_Init();

    PORTC = 0;
    PORTD = 0;
    PORTE = 0;

```

```

Lcd_Init();
Lcd_Cmd(_LCD_CLEAR);
Lcd_Cmd(_LCD_CURSOR_OFF);
i=0;
while (1) {
    if (i==1){
Lcd_Cmd(_LCD_CLEAR);
Lcd_Out(1,1, "threshold 1");
Lcd_Out(2,1, "threshold 2  ");
delay_ms(2000);
i=0;
    }else{

        idle_state() ;
        delay_ms(100);
    }
}
}

```